Traffic Sign Recognition System Using Deep Learning

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Abstract— Traffic Sign Recognition (TSR) is an essential component of Intelligent Transportation Systems (ITS), Advanced Driver Assistance Systems (ADAS), and autonomous driving applications. Accurate traffic sign detection enables timely interpretation of traffic regulations and road conditions, thereby improving road safety and driving efficiency. However, traditional traffic sign recognition methods often struggle under challenging conditions such as poor illumination, adverse weather, motion blur, and varying viewing angles. This paper presents a deep learning-based traffic sign detection system using the YOLOv11n model as the proposed approach. The performance of YOLOv11n is compared with YOLOv8n and YOLOv10n using the Self-Driving Cars dataset provided by Roboflow, which contains 4,969 annotated street-scene images across 15 traffic sign classes, including traffic lights, stop signs, and speed limit signs. Experimental results demonstrate that YOLOv11n achieves the best detection performance, obtaining a Precision of 95.91%, Recall of 91.98%, mAP@50 of 96.76%, and mAP@50–95 of 83.35%. Furthermore, robustness analysis under different image conditions confirms that the proposed model maintains reliable detection performance in challenging environments. These results demonstrate that YOLOv11n is an effective and efficient solution for accurate real-time traffic sign detection.

Keywords— *Traffic Sign Recognition (TSR), Deep Learning, Object Detection, Intelligent Transportation Systems (ITS), Advanced Driver Assistance Systems (ADAS)*

1. INTRODUCTION

Traffic signs are one of the most vital sources of information on the road. Traffic signs give the necessary information to the drivers about speed limits, warning signs, actions that must be performed, and prohibitions, thus ensuring safe driving and proper traffic management. In many cases, traffic signs help to avoid road accidents [2]. However, fatigue, distraction, bad visibility, and adverse weather conditions affect the understanding of traffic signs and may cause road accidents [2].

The traffic sign recognition problem is still hard because of the variety of different traffic sign appearances depending on size, orientation, lighting, and the complexity of the background. Motion blur, shadows, occlusion, faded traffic signs, and different weather conditions complicate the recognition problem even more [5], [7]. Traditional approaches to traffic sign recognition use hand-crafted features like color, shape, edge, etc., which do not always show good performance in real-world environments.

In order to solve the problems mentioned above, this work proposes deep learning-based traffic sign detection system with YOLOv11n as the proposed model. The performance of YOLOv11n was compared with YOLOv8n and YOLOv10n using Self-Driving Cars dataset for evaluating the traffic sign detection accuracy and robustness in different image conditions.

Experimental results showed that YOLOv11n gives the best performance among the considered models. Proposed traffic sign detection system demonstrates high accuracy and robustness of traffic signs detection under different environmental conditions and can be used for ITS, ADAS, and autonomous driving tasks.

2. TRAFFIC SIGN IMAGE AND FEATURE ANALYSIS

It is not simply about recognizing various traffic signs when checking out pictures of traffic signs. It is important to identify the essential visual attributes to distinguish one traffic sign from another. This involves examining the form, color, symbols and text in the signs. Also, we should take into account the patterns and features that makes one traffic sign different from another. In the real world however, traffic sign images may be difficult to capture because they can be taken under poor illumination, under different weather conditions, from motion-blurred cameras, from shaded cameras, or from different angles. There may also be some traffic signs which are only partially visible or very small. Therefore, it's important that we're clever in the way we analyze these pictures and are able to identify strong features that will allow us to distinguish between them. For this purpose, we have been given a special tool called YOLOv11n that can easily detect and classify these features autonomously and allow us to accurately detect and classify the traffic signs in real time during road environments.

2.1 Analysis of Traffic sign image

Look at pictures of traffic signs, and you'll realize that they're all unique. Some are stop signs, some are speed limit signs, some are traffic lights and others are warning sign or mandatory signs. Not only do these traffic signs look different, they also have different shapes, colors, symbols and sizes. For example, the stop sign is red and octagonal, and the speed limit signs are circular, with varying numbers. Likewise, the traffic lights have various colors including red, yellow, green. The visual differences will enable the separation of one traffic sign from another.

In the real world, images of traffic signs may have issues like variations in lighting conditions, rain, fog, shadows, motion-blurring, and different vantage points. Also, some traffic signs can be hard to find because of their size, being partially visible, or having complex backgrounds. So, the object detection model needs to be trained to be robust enough to recognize objects in such a difficult context. The model can automatically extract the key performance values, such as shape, color, symbols and spatial patterns, which are important for it to recognize and classify traffic signs in real-road environments. [5], [7]

2.2 Motivation

The Current traffic sign recognition systems mainly focus on improving detection accuracy. They pay little attention to performance in real-world driving conditions. Factors like poor lighting, rain, fog, motion blur, changing viewing angles, and partial obstructions can greatly reduce detection reliability.

The aim of this work is to create a strong and accurate traffic sign detection system for real-world use. YOLOv11n is chosen as the proposed model and is compared with YOLOv8n and YOLOv10n to assess their detection performance and reliability under various image conditions. This comparison helps find the best model for real-time applications in Intelligent Transportation Systems (ITS), Advanced Driver Assistance Systems (ADAS), and autonomous driving.

2.3 Feature extraction

In this work, the YOLOv11n architecture is used for feature extraction, which automatically extracts discriminative visual features from the input traffic sign images. Its model is able to capture key features like shape, colour, edges and symbols to accurately detect and classify various traffic sign categories. The extracted features are then used to perform real-time efficient localization and classification of traffic signs in road environments.

2.3.1 Spatial Features**:** In the pictures of the traffic signs, there are important visual features such as edges, shapes, and boundaries, which will help us identify different traffic signs. These are known as spatial features. The YOLOv11n model employs convolutional layers which automatically identify these features in the input images. This assists in distinguishing the shapes of traffic signs, including circles, triangles, rectangles and octagons, which are essential for categorizing traffic signs. The filters are applied to all parts of the image so that the model may know where and what traffic signs are. Therefore, the model can correctly recognize and classify the traffic signs under different positions, sizes, and angles.

$$F(i,j)=m\sum n\sum I(i+m,j+n)\cdot K(m,n) \quad (1)$$

Where:

- $F(i,j)$ = Output feature map at pixel location (i,j)
- $I(i,j)$ = Input traffic sign image
- $K(m,n)$ = Convolution kernel (filter)
- m,n = Kernel dimensions

2.3.2 Shape Features: Shape is one of the most critical attributes used for traffic sign recognition since traffic signs have standardized geometric shapes; for example, stop signs are octagonal, speed limit signs are circles, warning signs are triangles, and informational signs are rectangular. These unique shapes enable the differentiation of the traffic sign categories from each other.

2.3.3 Color Features: Colors are important for traffic sign recognition because different colors are used in traffic signs to signify different things. For example, stop signs are red, speed limit signs usually have red borders with white backgrounds, while traffic lights use red, yellow, and green colors to indicate different actions. The YOLOv11n model learns these color patterns during training when the model is trained with a large amount of traffic sign images. The model can be used to accurately classify various traffic sign categories by using color information along with other visual attributes, like shape and symbols. This leads to better detection and classification, even under varying light and road conditions of traffic signs.

2.3.4 Edge features: Edge attributes include the boundaries and contours of the traffic sign. Clear edge detection assists in separating the traffic sign from the background, especially when detecting traffic signs in complex road scene. The algorithm of edge detection may be illustrated by the following equation:

$$\mathrm{Gx} = \frac{\partial \mathrm{x}}{\partial \mathrm{I}} \times \frac{\partial \mathrm{y}}{\partial \mathrm{I}} \quad (2)$$

where:

- I = Input traffic sign image
- Gx = Horizontal edge gradient
- Gy= Vertical edge gradient

2.3.5 Symbol Features: Symbol attributes are the visual information contained within traffic signs including numbers, arrows, traffic light, pedestrians, and others. This is the information which defines the meaning of a particular traffic sign and distinguishes traffic signs having similar shapes or colors.

2.3.6 Multi-scale Features: Different sizes of traffic signs are caused by the varying distances from the traffic sign to the camera. Traffic signs closer to the vehicle will be larger, while more distant signs will be significantly smaller. Thus, the recognition algorithm should be scale invariant.

3. RELATED WORK

A study by W. E. Elside and A. J. Abougarair [1] proposed a Convolutional Neural Network (CNN)-based traffic sign recognition system for classifying traffic signs from road images. The study demonstrated that CNNs can automatically learn visual features and achieve high classification accuracy compared to traditional image processing methods. However, the work mainly focuses on traffic sign classification and does not evaluate modern object detection architectures such as YOLO for real-time traffic sign detection.

Y. Zhu and W. Q. Yan [2] developed a deep learning approach to traffic sign recognition which leverages the power of convolutional neural networks in traffic sign classification. In the paper, they addressed image processing, feature extraction, and classification methods with benchmark traffic sign datasets. It is evident from the experiment conducted that deep learning provides much better recognition rates than traditional methods. However, the work primarily emphasizes the issue of classification efficiency without comparing lightweight object detection approaches.

An Incremental Improvement, J. Redmon and A. Farhadi [3] propose an improved version of the YOLO architecture, which is capable of real-time object detection with high accuracy and object detection at different scales. The authors demonstrate that the new model achieves great speed and improved localization performance compared to the previous versions of YOLO. Nevertheless, the paper represents the general object detection framework, and there is no research on the applicability of the proposed framework for traffic sign recognition.

Deep learning approach to traffic sign recognition based on Convolutional Neural Networks was introduced by Z. Qin and W. Q. Yan [4]. It was shown how deep learning approach is efficient for automatic feature extraction and classification from traffic signs images. The study illustrates the efficiency of using CNNs for traffic sign recognition in contrast to handcrafted feature extraction methods. The study does not consider object detection approaches like YOLO.

R. K. Megalingam et al. [5] designed an Indian Traffic Sign Detection and Recognition System based on deep learning algorithms. This system attempted to overcome the problems involved in recognizing Indian traffic signs under diverse road conditions and exhibited potential detection accuracy. In this study, the role of deep learning was highlighted in relation to the intelligent transportation systems. Yet, the work is limited to Indian traffic signs and lacks a comparative study of recently designed lightweight YOLO architectures.

G. K. Samdariya and D. K. Meda [6] designed a deep learning approach to detect and recognize Indian traffic signs to enhance road safety and aid intelligent transportation systems. The authors successfully proved that deep learning techniques can help in detecting traffic signs in complex road environments. Nevertheless, a comparative study of existing state-of-the-art object detection architectures is not adequately covered in this paper.

A. A. Alshammari [7] proposes traffic sign recognition technique utilizing the YOLOv8 framework. The authors prove that YOLOv8 allows for much faster detection with higher accuracy than previous models which makes it more appropriate for self-driving cars applications. Nevertheless, the paper focuses only on evaluating the performance of YOLOv8 and does not present any comparative analysis with new generations of YOLO models, i.e., YOLOv10 and YOLOv11.

B. Xie and W. X. Xiong [8] introduce a real-time embedded traffic sign recognition system implemented via efficient convolutional neural network. The authors state that their system allows for high recognition rate with lower computational costs which makes it more appropriate for embedded systems. Nevertheless, no recent object detection frameworks are evaluated in the context of real-time traffic sign recognition task.

P. S. Zaki et al. [9] designed an innovative deep learning-based traffic sign detection and recognition framework that can detect and recognize traffic signs in road environment. The proposed method obtained high detection accuracy results and proved that deep learning can be used in intelligent transport systems. But the research does not include comparing several lightweight object detection frameworks and does not provide analysis of their efficiency depending on traffic conditions.

C. Kim et al. [10] suggested a deep learning-based real-time traffic sign recognition framework for urban areas. The suggested framework was mainly concerned about improving recognition results under realistic driving conditions. Thus, reliable real-time detection of traffic signs was proven. However, the research is mainly devoted to a single deep learning framework and does not include comprehensive comparative study of recent YOLO-based approaches.

4. Proposed Method

The proposed method uses a deep learning object detection framework for traffic sign detection. Three pre-trained YOLO models—YOLOv8n, YOLOv10n, and YOLOv11n—were trained and evaluated with the same dataset and experimental settings. The analysis showed that YOLOv11n had the best overall performance, so it was chosen as the proposed model. The system was also evaluated with standard detection metrics and tested under simulated conditions using Night Vision, Fog, Rain, and Gaussian Blur filters to check its robustness.

4.1 Flow of Proposed Model.

The workflow of the proposed traffic sign detection system is shown in Fig. X. First, the team collects the traffic sign dataset and preprocesses it by resizing images, checking annotations, and organizing the data for training, validation, and testing. They then use data augmentation techniques to increase dataset diversity and improve the model’s ability to generalize.

The preprocessed images are input into the pre-trained YOLOv11n model for training. Once training is complete, the model detects traffic signs by predicting the bounding box, class label, and confidence score for each object in the input image. Finally, the proposed model is evaluated using Precision, Recall, mAP@50, and mAP@50-95. To test its strength under tough conditions, the trained model undergoes additional tests with simulated environmental filters, which include Night Vision, Fog, Rain, and Gaussian Blur.

4.2 Pre-trained model

Pre trained models are deep learning models that have been already trained on huge datasets and can be adjusted for a specific job via transfer learning, kind of like reusing knowledge. When you start from pre-trained weights the network can pick up more generalized visual patterns faster so the fitting phase is shorter and the detection results tend to be better even if you only have a smaller number of training samples. In this work, the pre-trained YOLO versions were finetuned using the Self-Driving Cars dataset, for the purpose of traffic sign detection. With transfer learning the models are able to learn the distinctive traits of traffic signs, like their traffic sign characteristics, including shape, color, and surface texture and this leads to more reliable object localization and category recognition.

***4.2.1 YOLOv8n*:** YOLOv8n is a lightweight object detection model designed to achieve a balance between detection accuracy and computational efficiency. It utilizes a single-stage detection framework that predicts object locations and class labels simultaneously, enabling real-time performance. The model incorporates an anchor-free detection head and an improved backbone architecture for effective feature extraction at multiple scales. In this study, YOLOv8n was fine-tuned using pre-trained weights on the Self-Driving Cars dataset to establish a baseline for comparison with YOLOv10n and the proposed YOLOv11n model.

Despite its fast inference speed and low computational complexity, YOLOv8n may experience reduced detection accuracy for small or partially occluded traffic signs and under challenging environmental conditions, making it less robust than more advanced YOLO architectures.

4.2.2 YOLOv10n: YOLOv10n is an enhanced single-stage object detection algorithm which aims at improving detection accuracy without compromising on real-time detection. YOLOv10n includes architectural improvements and detection head that are optimized for efficient feature learning and low computational cost. YOLOv10n is able to detect multiple scales of features which facilitates detection of traffic signs of different shapes and sizes. In the current research, YOLOv10n has been fine-tuned with pre-trained weights on the Self-Driving Cars dataset and tested for comparison with YOLOv8n and proposed YOLOv11n models.

Even though YOLOv10n can provide enhanced feature extraction and reduced computational complexity, it may suffer poor detection results in case of heavy occlusion and drastic changes in illumination as well as complicated roads scenes.

4.2.3 YOLOv11n: YOLOv11n is the most recent lightweight object detection network from the family of YOLO networks, which is developed for the detection of objects with high accuracy while retaining real-time speed. The network uses enhanced backbone, neck, and detection head that help learn richer multi-scale features so as to accurately localize and classify traffic signs of different sizes with complex backgrounds. YOLOv11n was fine-tuned using the weights pretrained on the Self-Driving Cars dataset using the approach of transfer learning in this study. The network makes predictions regarding the coordinates of the bounding box, the object's confidence, and the class label at once, thereby making it suitable for the real-time detection of traffic signs. Due to its efficient architecture and rich feature learning capabilities, YOLOv11n has been chosen as the proposed network for this research.

Even though YOLOv11n has performed well in this study, it is possible that its detection performance can be affected in extremely difficult circumstances.

4.3 Evaluation measures of proposed

The performance of the proposed YOLOv11n model was evaluated using standard object detection metrics, namely Precision, Recall, mAP@50, and mAP@50–95. These metrics assess the model's ability to accurately detect and localize traffic signs under different driving conditions.

Precision measures the proportion of correctly detected traffic signs among all predicted detections and indicates the model's ability to minimize false positive predictions.

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

Where TP represents True Positives and FP represents False Positives.

Recall measures the proportion of correctly detected traffic signs among all actual traffic signs present in the dataset. It evaluates the capability of the model to identify all relevant objects.

$$Recall = \frac{TP}{TP + FN} \tag{4}$$

where FN denotes False Negatives.

The Mean Average Precision at IoU 0.50 (mAP@50) evaluates the average detection performance when the predicted bounding box overlaps the ground truth by at least 50%. It is calculated as the mean of the Average Precision (AP) values over all traffic sign classes.

$$IoU = \frac{Area\ of\ Overlap}{Area\ of\ Union} \tag{5}$$

Where:

- Area of Overlap = Common area between the predicted box and the ground truth box.
- Area of Union = Total area covered by both boxes combined.

$$\text{mAP@50} = \frac{1}{N}\sum_{i=1}^{N} AP_i \tag{6}$$

where N is the total number of traffic sign classes and AP_i is the Average Precision of the i^th class.

The Mean Average Precision at IoU 0.50–0.95 (mAP@50–95) provides a more comprehensive evaluation by averaging the Average Precision over multiple IoU thresholds ranging from 0.50 to 0.95 with an interval of 0.05. This metric evaluates both detection accuracy and localization quality under stricter matching conditions.

$$\text{mAP@50} - 95 = \frac{1}{10}\sum_{i=0}^{9} AP_{0.50+0.05i} \tag{7}$$

where:

- i = 0,1,…,9
- $AP_{0.50+0.05i}$ is the Average Precision calculated at each IoU threshold from 0.50 to 0.95 with an interval of 0.05.

Mean Average Precision(mAP): It is the main evaluation criterion for object detection algorithms. This metric evaluates the object detection performance using both precision and recall of all classes of traffic signs. In this study, mAP@50 and mAP@50-95 are adopted to assess the detection accuracy of the proposed models.

Validation Loss: It is the metric that determines the error of the model with respect to the validation set during the training process.

4.4 YOLOv11n Detection and Classification

The YOLOv11n model is designed for traffic sign detection and classification in a single forward pass. Following preprocessing and data argumentation, the input image is fed to the YOLOv11n network and then a deep convolutional feature extractor that extracts a feature pyramid comprising hierarchically layered features encoding shape, color, and structure of the traffic signs, through multiple levels. The extracted features at different scales are integrated by the neck which is able to aggregate information from various receptive fields, helping the detection of traffic signs with varying scales.

The detection head predicts the target object location (box coordinates), object scores, and traffic sign classes of detected objects simultaneously. During inference time, Non-

Maximum Suppression (NMS) is used to remove duplicated bounding box proposals by only selecting candidates with higher confidence. Then the predicted results include a traffic sign, its position, the object confidence value, and the traffic sign class. This end-to-end framework achieves robust and accurate traffic sign detection in real-time scenarios.

5. EXPERIMENTS AND RESULTS

5.1 Software and setup tools

The proposed traffic sign detection system was developed using Python 3.12 in the Visual Studio Code (VS Code) environment. The YOLOv8n, YOLOv10n, and YOLOv11n models were implemented using the Ultralytics YOLO framework with the PyTorch 2.12.1 deep learning library. Image preprocessing, augmentation, and visualization were performed using OpenCV, NumPy, Matplotlib, and Pandas. All experiments were conducted on a system running Windows 11 (64-bit) equipped with an NVIDIA GeForce RTX 3050 Laptop GPU (6 GB VRAM) and CUDA 12.6 to accelerate model training and inference. The Self-Driving Cars dataset, consisting of 15 traffic sign classes, was used for training, validation, and performance evaluation of the proposed model.

5.2 Experimental dataset

The dataset used in the current research project is the Self-Driving Cars dataset downloaded from Roboflow Universe. The dataset includes 4,969 labeled street scene images having 15 types of traffic signs like Red Light, Green Light, Stop Sign, Speed Limit 100, Speed Limit 110, Speed Limit 120, among others. All the traffic signs are annotated with bounding boxes, making it possible to detect objects and traffic signs.

All images are preprocessed and resized to meet the requirements for the input size of the YOLOv8n, YOLOv10n, and YOLOv11n models. The Roboflow divides all images automatically into three parts (train, validation, and test set). The labeling and organization of the data make it possible to train models to recognize traffic signs in different conditions.

5.3 Training and testing phase

The YOLOV11nmodelwas trained on the Self-Driving Cars dataset by utilizing pre-trained weights using the method of transfer learning. To make the model to effectively learn training and tune its parameters, as well as to test it unbiasedly and effectively, the dataset was divided into training (71.04%), validation (16.12%), and testing (12.84%) sets accordingly. For training: Image preprocessing – The input images were resized to640×640.

Data augmentation – Ultralytics framework data augmentation technique for training images.

The dataset was trained with batch size=16 and 50 epochs. The framework automatically chose the appropriate optimizer. During the entire training of data, the model will detect and identify the traffic signs by adjusting the loss of localization and classification. Each epoch was terminated with a validation dataset, which also auto saved the model if a better result is achieved and trained. For testing: After the completion of training, the best SAVED model is tested and evaluated by unseen data testing the model was evaluated with mAP, precision, recall, and mAP@50, mAP@50:95.

5.4 Result comparision and discussion

Table 1: Performance Comparison of YOLO Models

Model	Precision	Recall	mAP@50	mAP@50-95
YOLOv8n	89.36	87.26	92.06	78.90
YOLOv10n	93.60	89.90	95.10	82.70
YOLOv11n	95.91	91.98	96.76	83.35

A comparison of the three YOLO models is given in Table 1. The table shows that YOLOv8n obtained the worst performance of all the models, with a Precision of 89.36%, a Recall of 87.26% an mAP@50 of 92.06%, and an mAP@50–95 of 78.90%. Even though its detection accuracy is acceptable, its localization performance is comparatively lower than that of the other models.

YOLOv10n showed improved detection performance which reported a Precision of 93.60%, Recall of 89.90%, mAP@50 of 95.10% and mAP@50–95 of 82.70%. We saw that YOLOv10n does better in terms of object localization and overall detection performance when compared to YOLOv8n also it preserved efficiency in inference.

In terms of all the models evaluated YOLOv11n did the best which reported a Precision of 95.91%, Recall of 91.98%, mAP@50 of 96.76% and mAP@50–95 of 83.35%. The results which we got are that YOLOv11n does better in terms of traffic sign detection accuracy and also has better localization performance than the rest of the models. Thus, YOLOv11n was put forth as the traffic sign detection model of choice due to its improved accuracy, robustness and overall performance.

Table 2: Detection Confidence Comparison under Different Image Conditions

Image Condition	YOLOv8n	YOLOv10n	YOLOv11n
Original	86.50	89.74	91.20
Night Vision	91.72	88.67	95.27
Fog	80.53	94.91	94.32
Rain	54.93	92.32	49.84
Gaussian Blur	81.16	93.63	91.05

Table 2 shows the robustness analysis of the three YOLO models subjected to various image transformations, including the Original, Night Vision, Fog, Rain, and Gaussian Blur. The evaluation criteria used to analyze the performance of the models include the detection confidence (%) of the Speed Limit 80 traffic sign.

From Table 2, it is noted that the three models performed well in detecting the original image with a high level of confidence. In particular, the proposed YOLOv11n model had the highest confidence level (91.20%) compared to YOLOv10n (89.74%) and YOLOv8n (86.50%). This means that the proposed model has learned more discriminative features for effective detection of traffic signs under normal lighting conditions.

In terms of Night Vision filter, the performance of the proposed YOLOv11n model improved further by detecting the speed limit traffic sign with 95.27% confidence level. Additionally, the YOLOv8n confidence improved to 91.72%. However, there was a reduction in the confidence of YOLOv10n (88.67%) under the same lighting condition. It means that the model proposed has better performance than other models in low-lighting conditions.

Concerning the Fog filter, YOLOv10n and YOLOv11n were more robust to reduced visibility due to foggy environments since they had high confidence levels (94.91% and 94.32%, respectively). On the contrary, there was a reduction in confidence levels of YOLOv8n (80.53%).

Among all filters applied, Rain filter made the most significant impact on the detection results. YOLOv8n identified the traffic sign as Speed Limit 60 with a confidence rate of 54.93%, whereas YOLOv11n correctly identified the traffic sign with a decreased confidence rate of 49.84%. In contrast to that, YOLOv10n demonstrated the greatest confidence rate of 92.32% despite the presence of a simulated rain filter in this experiment.

In case of Gaussian Blur (21 × 21), all models were capable of detecting the traffic sign, and the confidence rates of YOLOv10n and YOLOv11n remained high. Thus, YOLOv10n reached the highest confidence level of 93.63%, followed by YOLOv11n with the confidence rate of 91.05% and YOLOv8n with 81.16% confidence rate. It can be seen that the influence of moderate blur on detection capability of YOLOv10n and YOLOv11n models is lower compared to that of YOLOv8n.

In general, YOLOv11n had the most stable detection performance on most images and had the highest comprehensive evaluation score. Therefore, YOLOv11n was chosen as the proposed model with higher accuracy and good generalization ability.

6. Conclusion

In this study, the YOLOv11n architecture is used as the proposed method for detecting traffic signs. We compared the performance of YOLOv11n with YOLOv8n and YOLOv10n on the Self-Driving Cars dataset with 15 classes of traffic signs. We evaluated the proposed model with standard object detection metrics such as Precision, Recall, mAP@50, and mAP@50–95 metrics.

YOLOv11n achieves a Precision score of 95.91%, Recall score of 91.98%, mAP@50 of 96.76%, and mAP@50–95 score of 83.35%. This proves that our proposed model has better detection accuracy and localization ability than YOLOv8n and YOLOv10n models.

To evaluate the performance of our proposed model under various image changes, we tested it on Night Vision, Foggy, Rainy, and Gaussian Blur images. As shown in Figure 11, images that were subjected to Night Vision, Rain, and Gaussian Blur image transformations saw a slight decrease in confidence, but YOLOv11n still delivered consistent detection results for most images which signifies our model has excellent generalization performance for real-world traffic signs.

YOLOv11n's detector shows strong performance in detecting traffic signs in real-time which makes it ideal for ITS and ADAS applications.

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